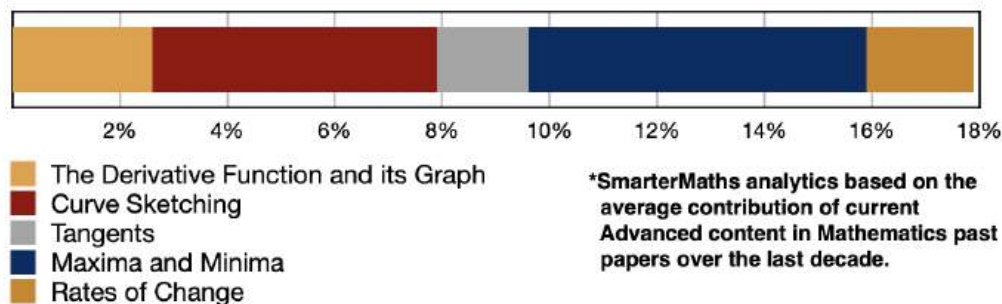


ADV: Calculus (Adv), C3 Applications of Calculus (Adv) The Derivative Function and its Graph (Y12)

Teacher: Ljiljana Todic

Exam Equivalent Time: 70.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

C3 Applications of Calculus



HISTORICAL CONTRIBUTION

- *C3 Applications of Calculus* is the biggest topic in the Advanced course, contributing a massive 17.9% to past Mathematics exams, on average over the past decade.
- This topic has been split into five sub-topics for analysis purposes: 1-*The Derivative Function and its Graph* (2.6%), 2-*Curve Sketching* (5.3%), 3-*Tangents* (1.7%), 4-*Maxima and Minima* (6.3%) and 5-*Rates of Change* (2.0%).
- This analysis looks at *The Derivative Function and its Graph*.

HSC ANALYSIS - What to expect and common pitfalls

- *The Derivative Function and its Graph* is a challenging sub-topic that demands a solid conceptual understanding of the first and second derivative.
- Students are regularly asked to explore the graphical relationship between $f(x)$ and $f'(x)$. This question type has caused problems in the past and deserves attention.
- This sub-topic has been tested 7 times in the last decade and has produced sub-50% mean marks on a majority of occasions. A revision focus is recommended.
- The 2020 exam tested composite functions in this context for the first time, producing an 11% mean mark (lowest in the exam). This should be carefully reviewed (see 2020 Adv 10 MC).

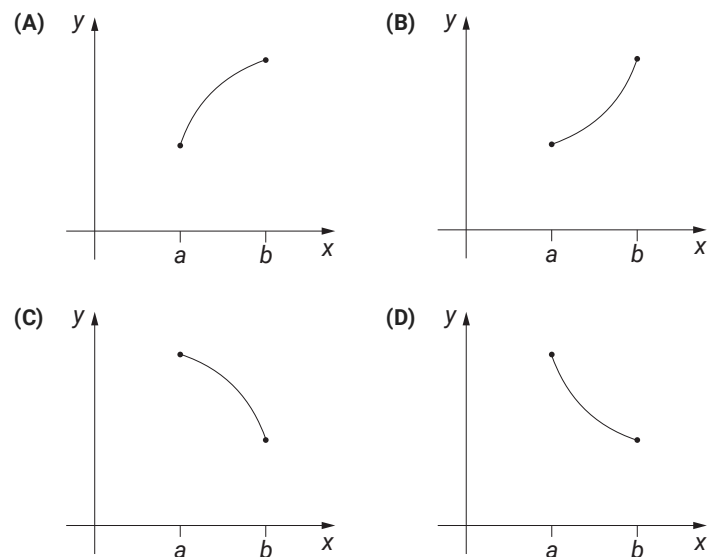
Questions

1. Calculus, 2ADV C3 2017 HSC 4 MC

The function $f(x)$ is defined for $a \leq x \leq b$.

On this interval, $f'(x) > 0$ and $f''(x) < 0$.

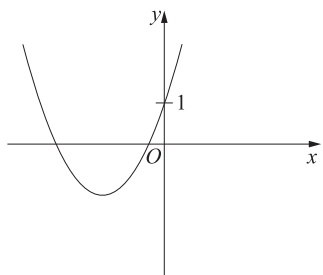
Which graph best represents $y = f(x)$?



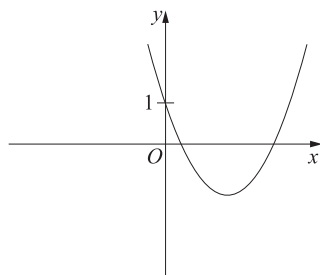
2. Functions, 2ADV F1 2020 HSC 5 MC

Which of the following could represent the graph of $y = -x^2 + bx + 1$, where $b > 0$?

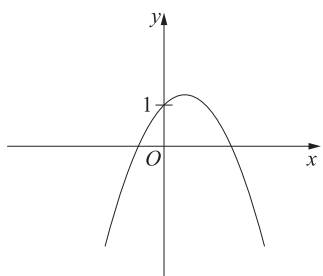
A.



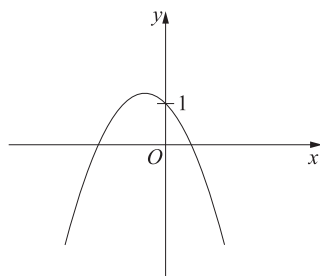
B.



C.

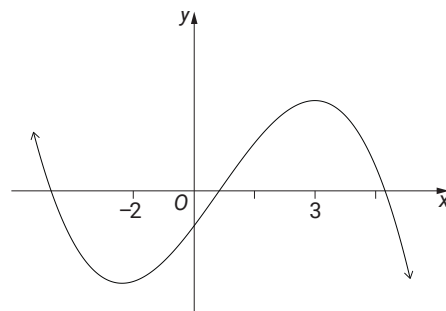


D.



3. Calculus, 2ADV C3 2021 HSC 7 MC

The diagram shows part of $y = f(x)$ which has a local minimum at $x = -2$ and a local maximum at $x = 3$.

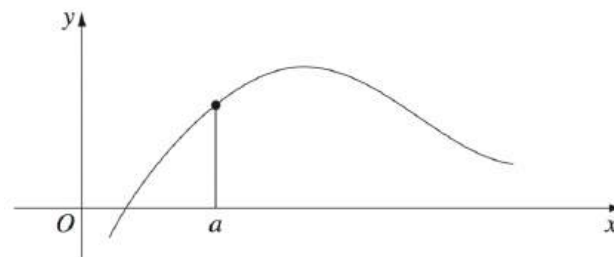


Which of the following shows the correct relationship between $f''(-2)$, $f(0)$ and $f'(3)$?

- A. $f(0) < f'(3) < f''(-2)$
- B. $f(0) < f''(-2) < f'(3)$
- C. $f''(-2) < f'(3) < f(0)$
- D. $f''(-2) < f(0) < f'(3)$

4. Calculus, 2ADV C3 2012 HSC 4 MC

The diagram shows the graph $f(x)$.

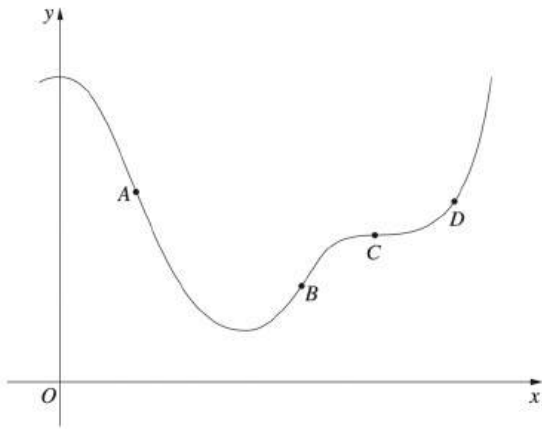


Which of the following statements is true?

- (A) $f'(a) > 0$ and $f''(a) < 0$
- (B) $f'(a) > 0$ and $f''(a) > 0$
- (C) $f'(a) < 0$ and $f''(a) < 0$
- (D) $f'(a) < 0$ and $f''(a) > 0$

5. Calculus, 2ADV C3 2013 HSC 8 MC

The diagram shows points **A**, **B**, **C** and **D** on the graph $y = f(x)$.

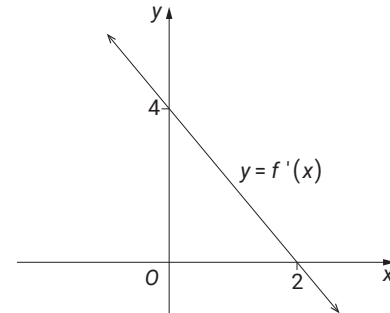


At which point is $f'(x) > 0$ and $f''(x) = 0$?

- (A) **A**
 - (B) **B**
 - (C) **C**
 - (D) **D**
-

6. Calculus, 2ADV C3 2017 HSC 9 MC

The graph of $y = f'(x)$ is shown.



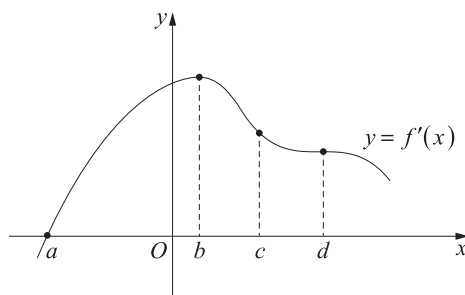
The curve $y = f(x)$ has a maximum value of 12.

What is the equation of the curve $y = f(x)$?

- (A) $y = x^2 - 4x + 12$
 - (B) $y = 4 + 4x - x^2$
 - (C) $y = 8 + 4x - x^2$
 - (D) $y = x^2 - 4x + 16$
-

7. Calculus, 2ADV C3 2018 HSC 9 MC

The diagram shows the graph of $f'(x)$, the derivative of a function.

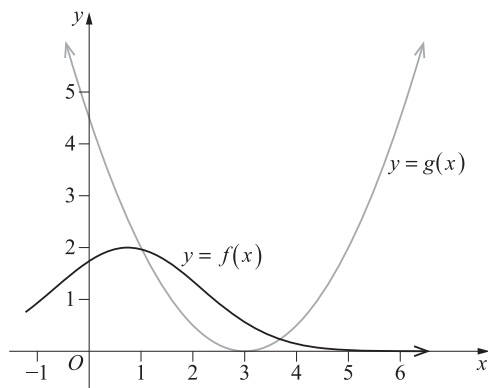


For what value of x does the graph of the function $f(x)$ have a point of inflexion?

- (A) $x = a$
- (B) $x = b$
- (C) $x = c$
- (D) $x = d$

8. Calculus, 2ADV C3 2020 10 MC

The graph shows two functions $y = f(x)$ and $y = g(x)$.



Define $h(x) = f(g(x))$.

How many stationary points does $y = h(x)$ have for $1 \leq x \leq 5$?

- A. 0
- B. 1
- C. 2
- D. 3

9. Calculus, 2ADV C3 2013 HSC 12a

The cubic $y = ax^3 + bx^2 + cx + d$ has a point of inflexion at $x = p$.

Show that $p = -\frac{b}{3a}$. (2 marks)

10. Calculus, 2ADV C3 2011 HSC 4c

The gradient of a curve is given by $\frac{dy}{dx} = 6x - 2$. The curve passes through the point $(-1, 4)$.

What is the equation of the curve? (2 marks)

11. Calculus, 2ADV C3 2014 HSC 11f

The gradient function of a curve $y = f(x)$ is given by $f'(x) = 4x - 5$. The curve passes through the point $(2, 3)$.

Find the equation of the curve. (2 marks)

12. Calculus, 2ADV C3 2014 HSC 14a

Find the coordinates of the stationary point on the graph $y = e^x - ex$, and determine its nature. (3 marks)

13. Calculus, 2ADV C4 2008 HSC 5a

The gradient of a curve is given by $\frac{dy}{dx} = 1 - 6 \sin 3x$. The curve passes through the point $(0, 7)$.

What is the equation of the curve? (3 marks)

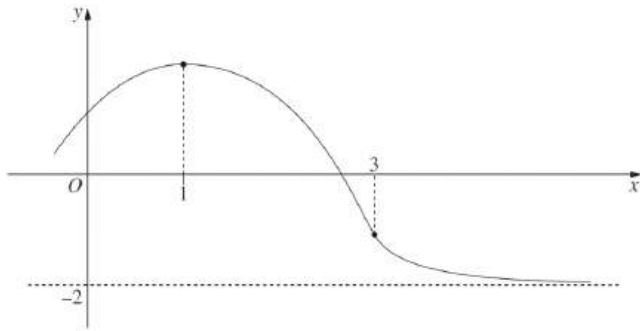
14. Calculus, 2ADV C3 2010 HSC 8d

Let $f(x) = x^3 - 3x^2 + kx + 8$, where k is a constant.

Find the values of k for which $f(x)$ is an increasing function. (2 marks)

15. Calculus, 2ADV C3 2011 HSC 9c

The graph $y = f(x)$ in the diagram has a stationary point when $x = 1$, a point of inflexion when $x = 3$, and a horizontal asymptote $y = -2$.

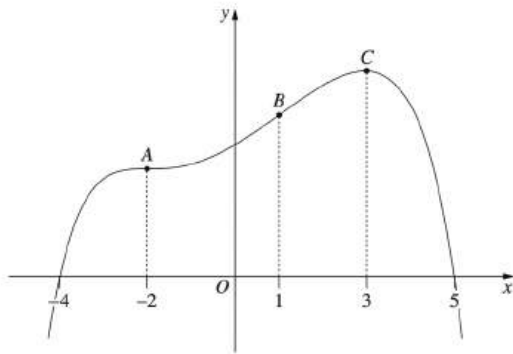


Sketch the graph $y = f'(x)$, clearly indicating its features at $x = 1$ and at $x = 3$, and the shape of the graph as $x \rightarrow \infty$. (3 marks)

16. Calculus, 2ADV C3 2014 HSC 14e

The diagram shows the graph of a function $f(x)$.

The graph has a horizontal point of inflexion at A , a point of inflexion at B and a maximum turning point at C .



Sketch the graph of the derivative $f'(x)$. (3 marks)

17. Calculus, 2ADV C3 2018 HSC 14c

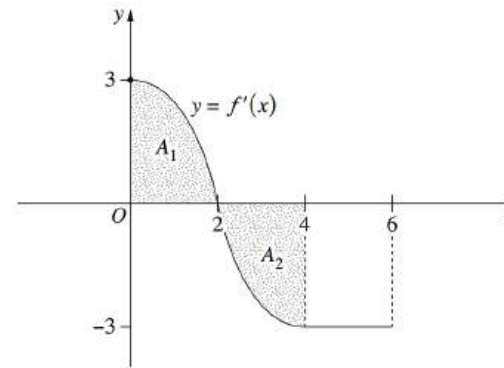
Let $f(x) = x^3 + kx^2 + 3x - 5$, where k is a constant.

Find the values of k for which $f(x)$ has NO stationary points. (3 marks)

18. Calculus, 2ADV C3 2010 HSC 9b

Let $y = f(x)$ be a function defined for $0 \leq x \leq 6$, with $f(0) = 0$.

The diagram shows the graph of the derivative of f , $y = f'(x)$.



The shaded region A_1 has area 4 square units. The shaded region A_2 has area 4 square units.

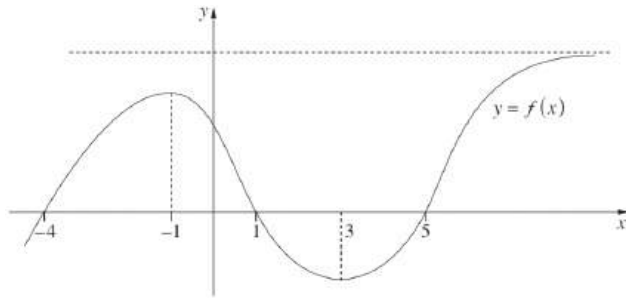
i. For which values of x is $f(x)$ increasing? (1 mark)

ii. What is the maximum value of $f(x)$? (1 mark)

iii. Find the value of $f(6)$. (1 mark)

iv. Draw a graph of $y = f(x)$ for $0 \leq x \leq 6$. (2 marks)

19. Calculus, 2ADV C3 2009 HSC 8a

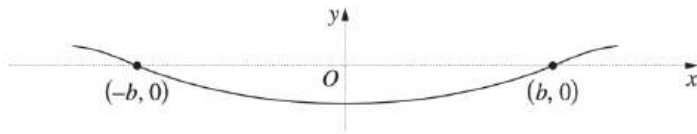


The diagram shows the graph of a function $y = f(x)$.

- For which values of x is the derivative, $f'(x)$, negative? (1 mark)
- What happens to $f'(x)$ for large values of x ? (1 mark)
- Sketch the graph $y = f'(x)$. (2 marks)

20. Calculus, 2ADV C4 2008 HSC 9c

A beam is supported at $(-b, 0)$ and $(b, 0)$ as shown in the diagram.



It is known that the shape formed by the beam has equation $y = f(x)$, where $f(x)$ satisfies

$$f''(x) = k(b^2 - x^2), \quad (k \text{ is a positive constant})$$

and $f'(-b) = -f'(b)$.

i. Show that $f'(x) = k\left(b^2x - \frac{x^3}{3}\right)$. (2 marks)

- ii. How far is the beam below the x -axis at $x = 0$? (2 marks)

21. Calculus, 2ADV C3 2021 HSC 16

For what values of x is $f(x) = x^2 - 2x^3$ increasing? (3 marks)

Worked Solutions

1. Calculus, 2ADV C3 2017 HSC 4 MC

Interpreting $f'(x) > 0$,

$\Rightarrow$ gradient is always positive

Interpreting $f''(x) < 0$,

$\Rightarrow$ Curve is concave down

$\Rightarrow A$

2. Functions, 2ADV F1 2020 HSC 5 MC

$$y = -x^2 + bx + 1$$

$$\frac{dy}{dx} = -2x + b$$

S.P. occurs when $\frac{dy}{dx} = 0$:

$$-2x + b = 0$$

$$x = \frac{b}{2}$$

Since $b > 0$, SP occurs when $x > 0$

$\Rightarrow C$

3. Calculus, 2ADV C3 2021 HSC 7 MC

$$f''(-2) > 0 \quad (\text{concave up at } x = -2)$$

$$f(0) < 0 \quad (\text{see graph})$$

$$f'(3) = 0 \quad (\text{S.P.})$$

$$\therefore f(0) < f'(3) < f''(-2)$$

$\Rightarrow A$

4. Calculus, 2ADV C3 2012 HSC 4 MC

At $x = a$,
 $f'(a) > 0 \quad \therefore$ Cannot be C or D
 $f''(a) < 0$ because $f(x)$ is concave down at $x = a$
 $\therefore$ Cannot be B
 $\Rightarrow A$

5. Calculus, 2ADV C3 2013 HSC 8 MC

At A , $f'(x) < 0$
 At C , $f'(x) = 0$
 $\therefore$ It cannot be A or C .
 At D , $f''(x) > 0$ (concave up)
 At B , $f''(x) = 0$ (concavity changes)
 $\Rightarrow B$

◆ Mean mark 48%

6. Calculus, 2ADV C3 2017 HSC 9 MC

Find the equation of $f'(x)$:
 $m = -2$, $y\text{-int} = 4$
 $y = -2x + 4$

$$f(x) = \int -2x + 4 \, dx$$

$$= -x^2 + 4x + c$$

 Maximum $f(x) = 12$ when $f'(x) = 0$
 $-2x + 4 = 0$
 $x = 2$
 Substitute $x = 2$ into $f(x)$
 $\therefore 12 = -2^2 + 4 \cdot 2 + c$
 $c = 8$
 $\therefore f(x) = 8 + 4x - x^2$
 $\Rightarrow C$

7. Calculus, 2ADV C3 2018 HSC 9 MC

P.I. will occur when the graph of $f'(x)$
 has a turning point.
 $\therefore x = b$
 $\Rightarrow B$

◆ Mean mark 41%.

8. Calculus, 2ADV C3 2020 10 MC

$h(x) = f(g(x))$
 $h'(x) = g'(x) \times f'(g(x))$

◆◆ Mean mark 11%.

S.P.'s occur when $g'(x) = 0$ or $f'(g(x)) = 0$
 $g'(x) = 0$ when $x = 3$ (from graph)
 $f'(x) = 0$ when $x \approx 1$ (i.e. x -value is just under 1)

Find values of x when $g(x) \approx 1$:

By inspection, there are 2 values where

$g(x) \approx 1$, $x \in [1, 5]$

$\therefore$ There are 3 S.P.'s for $y = h(x)$, $x \in [1, 5]$
 $\Rightarrow D$

9. Calculus, 2ADV C3 2013 HSC 12a

Show $p = -\frac{b}{3a}$

$$y = ax^3 + bx^2 + cx + d$$

$$y' = 3ax^2 + 2bx + c$$

$$y'' = 6ax + 2b$$

Given P.I. occurs when $x = p$

$\Rightarrow y'' = 0$ when $x = p$

$$\therefore 6ap + 2b = 0$$

$$6ap = -2b$$

$$p = -\frac{2b}{6a}$$

$$= -\frac{b}{3a} \quad \dots \text{ as required}$$

10. Calculus, 2ADV C3 2011 HSC 4c

$$\frac{dy}{dx} = 6x - 2$$

$$y = \int 6x - 2 \, dx$$
$$= 3x^2 - 2x + c$$

Since it passes through $(-1, 4)$,

$$4 = 3(-1)^2 - 2(-1) + c$$

$$4 = 3 + 2 + c$$

$$c = -1$$

$$\therefore y = 3x^2 - 2x - 1$$

11. Calculus, 2ADV C3 2014 HSC 11f

$$f'(x) = 4x - 5$$

$$f(x) = \int 4x - 5 \, dx$$
$$= 2x^2 - 5x + C$$

Given $f(x)$ passes through $(2, 3)$

$$3 = 2(2^2) - 5(2) + C$$

$$3 = 8 - 10 + C$$

$$C = 5$$

$$\therefore f(x) = 2x^2 - 5x + 5$$

12. Calculus, 2ADV C3 2014 HSC 14a

$$y = e^x - ex$$

$$\frac{dy}{dx} = e^x - e$$

$$\frac{d^2y}{dx^2} = e^x$$

$$\text{S.P. when } \frac{dy}{dx} = 0$$

$$e^x - e = 0$$

$$e^x = e^1$$

$$x = 1$$

$$\text{At } x = 1$$

$$y = e^1 - e = 0$$

$$\frac{d^2y}{dx^2} = e > 0 \Rightarrow \text{MIN}$$

$$\therefore \text{MINIMUM S.P. at } (1, 0)$$

13. Calculus, 2ADV C4 2008 HSC 5a

$$\frac{dy}{dx} = 1 - 6 \sin 3x$$

$$y = \int 1 - 6 \sin 3x \, dx$$
$$= x + 2 \cos 3x + c$$

Passes through $(0, 7)$

$$\Rightarrow 0 + 2 \cos 0 + c = 7$$

$$2 + c = 7$$

$$c = 5$$

$$\therefore \text{Equation is } y = x + 2 \cos 3x + 5$$

14. Calculus, 2ADV C3 2010 HSC 8d

$$f(x) = x^3 - 3x^2 + kx + 8$$

$$f'(x) = 3x^2 - 6x + k$$

$f(x)$ is increasing when $f'(x) > 0$

$$\Rightarrow 3x^2 - 6x + k > 0$$

$f'(x)$ is always positive

$\Rightarrow f'(x)$ is a positive definite.

i.e. when $a > 0$ and $\Delta < 0$

$$a = 3 > 0$$

$$\Delta = b^2 - 4ac$$

$$\therefore (-6)^2 - (4 \times 3 \times k) < 0$$

$$36 - 12k < 0$$

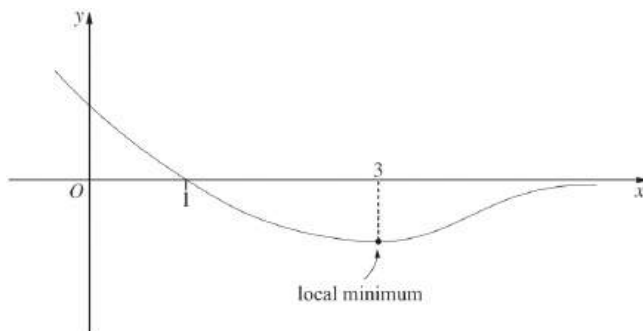
$$12k > 36$$

$$k > 3$$

$\therefore f(x)$ is increasing when $k > 3$.

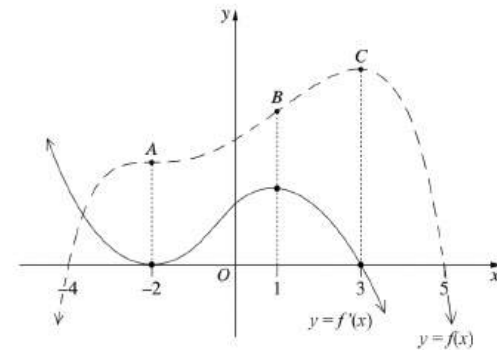
♦♦ Mean mark 28%.
MARKER'S COMMENT: The arithmetic required to solve $36 - 12k < 0$ proved the undoing of too many students in this question. TAKE CARE!

15. Calculus, 2ADV C3 2011 HSC 9c



♦ Mean mark 43%
IMPORTANT: Examiners *regularly* ask questions that require the graphing of an $f'(x)$ given the $f(x)$ graph and vice-versa. *KNOW IT!*

16. Calculus, 2ADV C3 2014 HSC 14e



17. Calculus, 2ADV C3 2018 HSC 14c

$$f(x) = x^3 + kx^2 + 3x - 5$$

$$f'(x) = 3x^2 + 2kx + 3$$

♦ Mean mark 49%.

No S.P.'s exist if $f'(x)$ has no roots,

$$\Delta < 0$$

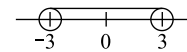
$$b^2 - 4ac < 0$$

$$(2k)^2 - 4 \times 3 \times 3 < 0$$

$$4k^2 - 36 < 0$$

$$k^2 - 9 < 0$$

$$(k - 3)(k + 3) < 0$$



$$\therefore -3 < k < 3$$

18. Calculus, 2ADV C3 2010 HSC 9b

- i. $f(x)$ is increasing when $f'(x) > 0$

From the graph

$f(x)$ is increasing when $0 \leq x < 2$

- ii. $f'(x) = 0$ when $x = 2$

$\therefore$ MAX at $x = 2$

$$\int_0^2 f'(x) dx = 4 \quad (\text{given since } A_1 = 4)$$

We also know

$$\begin{aligned} \int_0^2 f'(x) dx &= [f(x)]_0^2 \\ &= f(2) - f(0) \\ &= f(2) \quad (\text{since } f(0) = 0) \\ \Rightarrow f(2) &= 4 \end{aligned}$$

$\therefore$ MAX value of $f(x) = 4$

$$\begin{aligned} \text{iii. } \int_0^4 f'(x) dx &= A_1 - A_2 \\ &= 0 \end{aligned}$$

We also know

$$\begin{aligned} \int_0^4 f'(x) dx &= \int_2^4 f'(x) dx + \int_0^2 f'(x) dx \\ &= [f(x)]_2^4 + 4 \\ &= f(4) - f(2) + 4 \quad (\text{note } f(2) = 4) \\ &= f(4) \\ \Rightarrow f(4) &= 0 \end{aligned}$$

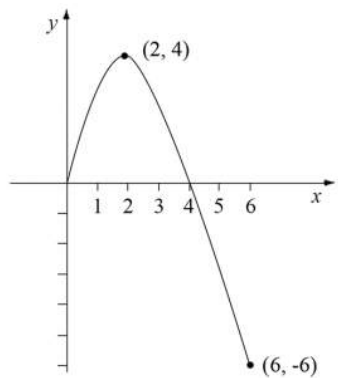
Gradient = -3 from $x = 4$ to $x = 6$

$$\begin{aligned} \therefore f(6) &= -3(6 - 4) \\ &= -6 \end{aligned}$$

- iv.

◆◆ Parts (ii) and (iii) proved particularly difficult for students with mean marks of 12% and 11% respectively.

◆◆ Mean mark 28%
EXAM TIP: Clearly identify THE



EXTREMES when given a defined domain. In this case, the origin is obvious graphically, and the other extreme at $x = 6$, is *CLEARLY LABELLED*!

19. Calculus, 2ADV C3 2009 HSC 8a

i. $f'(x) < 0$ when

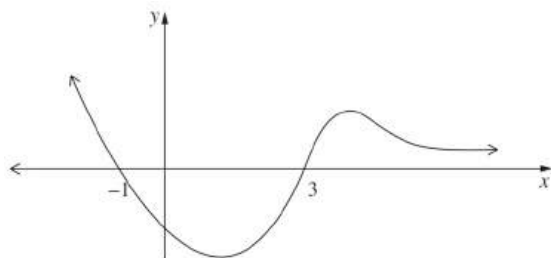
$$-1 < x < 3$$

ii. As $x \rightarrow \infty$

$$f'(x) \rightarrow 0$$

♦♦ Exact data not available.

iii.



♦♦ Exact data not available.

MARKER'S COMMENT: Poorly drawn graphs with axes not labelled and inaccurate scales were common.

i. Show $f'(x) = k\left(b^2x - \frac{x^3}{3}\right)$

$$f''(x) = k(b^2 - x^2)$$

$$f'(x) = \int k(b^2 - x^2) dx$$

$$= k \int b^2 - x^2 dx$$

$$= k\left(b^2x - \frac{x^3}{3}\right) + c$$

Since S.P. exists at $x = 0$

$$\Rightarrow f'(x) = 0 \text{ when } x = 0$$

$$0 = k(b^2 \cdot 0 - 0) + c$$

$$c = 0$$

$$\therefore f'(x) = k\left(b^2x - \frac{x^3}{3}\right) \text{ ... as required}$$

ii. $f(x) = \int f'(x) dx$

$$= k \int b^2x - \frac{x^3}{3} dx$$

$$= k\left(\frac{b^2x^2}{2} - \frac{x^4}{12}\right) + c$$

We know $f(x) = 0$ when $x = b$

$$\Rightarrow 0 = k\left(\frac{b^2 \cdot b^2}{2} - \frac{b^4}{12}\right) + c$$

$$c = -k\left(\frac{6b^4}{12} - \frac{b^4}{12}\right)$$

$$= -k\left(\frac{5b^4}{12}\right)$$

$$= -\frac{5kb^4}{12}$$

$\therefore$ When $x = 0$, the beam is $\frac{5kb^4}{12}$ units

below the x -axis.

21. Calculus, 2ADV C3 2021 HSC 16

$$f(x) = x^2 - 2x^3$$

$$f'(x) = 2x - 6x^2$$

$f(x)$ is increasing when $f'(x) > 0$

$$2x - 6x^2 > 0$$

$$2x(1 - 3x) > 0$$

$$x \in \left(0, \frac{1}{3}\right)$$